Recurrence of Greedy Valuation Walks and the OEIS A382357 Permutation Conjecture

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Abstract

We prove that the greedy sequence of distinct positive integers whose successive 2-adic valuations differ by one is a permutation of the positive integers, resolving the conjecture in OEIS A382357. More generally, for every integer base $b \geq 2$, every fixed valuation jump $d \geq 1$, and every finite legal seed, the resulting sequence lists every integer in the compatible valuation residue class. A last-exit argument converts hypothetical escape to large valuations into a contracting recurrence for positive integer edge-crossing counts. For the binary sequence we also prove the empirical pairing formula in A382360 and an exact five-term block around each new even power of two.

1 Definitions and the general statement

For an integer $b \geq 2$, define $v_b(n) = \max\{k \geq 0 : b^k \mid n\}$. This definition also applies when b is composite. Fix an integer $d \geq 1$ and a finite seed of distinct positive integers $a_1, \ldots, a_T$ satisfying

$$|v_b(a_{i+1}) - v_b(a_i)| = d \qquad (1 \leq i < T).$$

Thereafter choose a_{t+1} to be the smallest unused positive integer whose valuation differs from $v_b(a_t)$ by exactly d. There is always an available term, since the higher neighboring valuation class is infinite. All valuations belong to a single residue class $r \pmod d$, $0 \leq r < d$. Let

$$\mathcal{C}_{b,d,r} = \{n \geq 1 : v_b(n) \equiv r \pmod d\}.$$

Theorem 1 (Recurrence and exhaustive coverage)**.** *For every b, d and finite legal seed above, the full sequence consisting of the seed and its greedy continuation lists every member of $\mathcal{C}_{b,d,r}$ exactly once. In particular, with $b = 2$, $d = 1$, and $a_1 = 1$, it proves that OEIS A382357 is a permutation of the positive integers.*

The OEIS entry [1] records the permutation assertion as a conjecture. It already proves that every power of two appears. The argument below proves recurrence at every valuation level and therefore coverage of all integers in each class.

2 Queues, last exits, and a contraction

Normalize a valuation $r + dj$ to the level $j \geq 0$, so the sequence walks on the nonnegative integers by steps ± 1. Put $\lambda = b^d$ and

$$h_b(t) = t + \left\lfloor \frac{t-1}{b-1} \right\rfloor \qquad (t \geq 1).$$

These are, in increasing order, the positive integers not divisible by b. At level j the labels are $b^r \lambda^j h_b(t)$, with $t = 1, 2, \ldots$. Each selection uses the smallest unused label at that level. Thus, at a level absent from the seed, after c_j visits the next label is

$$Q_j = b^r \lambda^j h_b(c_j + 1). \tag{1}$$

Lemma 1 (Propagation of infinite visits)**.** *If one level is visited infinitely often, then every level is visited infinitely often, and every compatible integer appears.*

Proof. Suppose a level j is visited infinitely often and one neighboring level has only finitely many visits. The latter has a fixed smallest unused label M after its last visit. At every sufficiently late visit to j, the next chosen label is at most M. Distinctness permits only finitely many such choices, a contradiction. Infinite visits thus propagate across every edge of the level path. At any level, a fixed unused label would bound every subsequent selection from that level, again contradicting infinitely many distinct selections. □

Lemma 2 (Last-exit inequality)**.** *Suppose every level is visited only finitely many times. Let K be the greatest level appearing in the seed, and let U_j be the total number of upward crossings from j to $j+1$. Then every U_j with $j \geq K$ is a positive integer, and for every $k \geq K+2$,*

$$\lambda^2 U_k \leq U_{k-2} + U_{k-1} + \lambda^2 - \lambda^2/b - 1. \tag{2}$$

Proof. Since each finite set of levels receives finitely many visits, the level of the walk tends to infinity. For $j \geq K$, the walk starts below or on j and eventually stays above it. Thus the downward crossing count is $D_j = U_j - 1$, and $U_j \geq 1$. At a level $j > K$, which does not contain the initial term, every visit is an arrival; hence its total visit count is

$$c_j = U_{j-1} + D_j = U_{j-1} + U_j - 1. \tag{3}$$

Consider the final departure from level $k \geq K+2$. It must go upward: a downward departure would require a later crossing of k on the way to infinity. At this instant level $k-1$ has already had all its c_{k-1} visits. Level $k+1$ has had at least $U_k - 1$ visits, from the earlier upward crossings of this edge. Both levels are absent from the seed. The greedy comparison using (1) therefore gives

$$\lambda^2 h_b(U_k) < h_b(c_{k-1} + 1). \tag{4}$$

The inequality is strict because labels from distinct valuation classes are different.

Write $\alpha = b/(b-1)$. The explicit formula for h_b gives

$$h_b(t) \geq \alpha t - 1, \qquad h_b(c+1) \leq \alpha c + 1.$$

Consequently (4) and (3) imply

$$\begin{aligned} \lambda^2 U_k &< c_{k-1} + \frac{\lambda^2 + 1}{\alpha} \\ &= U_{k-2} + U_{k-1} + \lambda^2 - \lambda^2/b - 1/b. \end{aligned}$$

The quantities other than $1/b$ are integers, since $\lambda = b^d$. Rounding down gives (2). □

Proof of Theorem 1. If every level has finite visits, Lemma 2 applies. For a positive integer M, define

$$F(M) = \left\lfloor \frac{2M + \lambda^2 - \lambda^2/b - 1}{\lambda^2} \right\rfloor.$$

We have $F(M) \leq M - 1$: indeed

$$(\lambda^2 - 2)M - \lambda^2 + \lambda^2/b + 1 \geq \lambda^2/b - 1 > 0.$$

If two consecutive crossing counts are at most M, the next two are at most $F(M)$, by (2). Starting with $M = \max(U_K, U_{K+1})$, after at most M such pairs the positive crossing counts are bounded by zero. This contradiction proves that some level has infinitely many visits. Lemma 1 then gives every compatible integer, and distinctness is part of the rule. □

For the canonical binary sequence the central inequality reduces to

$$4U_k \leq U_{k-2} + U_{k-1} + 1.$$

Its incompatibility with an infinite positive integer sequence is the obstruction to escaping permanently to larger 2-adic valuations.

3 Paired powers of two and exact local blocks

Return to the canonical binary sequence $a(1), a(2), \ldots$. Let t_j be the index of the unique occurrence of 2^j. These indices are OEIS A382360 [2].

Theorem 2 (Pairing and a five-term block)**.** *For every $m \geq 0$,*

$$t_{2m+1} = t_{2m} + 1. \tag{5}$$

More precisely, for $m \geq 1$ and $x = 4^m$, with $t = t_{2m}$,

$$(a(t-1), a(t), a(t+1), a(t+2), a(t+3)) = (3x/2,\ x,\ 2x,\ 3x,\ 5x/2). \tag{6}$$

The initial block is $(a(1), a(2), a(3)) = (1, 2, 3)$.

Proof. The first visit to level j uses 2^j, because labels within that level are used in increasing order. Initially the walk visits levels $0, 1, 0$, with labels $1, 2, 3$.

Suppose the first visits to levels $2m-2$ and $2m-1$ are consecutive, and are followed by a return to level $2m-2$. The next visit to $2m-1$ is its second visit, with label $3 \cdot 2^{2m-1}$. At that time level $2m-2$ has been visited at least twice. Its next label is at least $5 \cdot 2^{2m-2}$, greater than the unused label $2^{2m} = 4 \cdot 2^{2m-2}$. Thus this second visit immediately enters the new level $2m$.

Put $x = 2^{2m}$. Level $2m-1$ has now been visited exactly twice. After choosing x, the lower candidate is $5x/2$ and the higher one is $2x$, so $2x$ is chosen. The next candidates are $3x$ and $4x$, so the walk returns to level $2m$, using $3x$. Its candidates are then $5x/2$ and $6x$, so $5x/2$ is chosen. This proves (6), supplies the consecutive new levels and immediate return needed for the next induction step, and establishes (5). All visits used in this induction exist by Theorem 1. □

The OEIS formula for A382360 is currently marked empirical; equation (5) proves it. The five-term identity gives additional local values without requiring a closed formula for the positions t_{2m}.

4 Reproducible checks

The proof is independent of finite computation. The supplied verifier compares a queue implementation with an independent direct scan for the smallest unused eligible integer, including arbitrary finite seeds and bases and jumps other than two and one. It checks the displayed OEIS prefix, twenty recorded inverse positions, the local blocks, and the crossing inequalities on finite first-arrival paths. Separate integer audits test the rounding and contraction estimates.

References

[1] R. Sigrist, OEIS A382357, https://oeis.org/A382357, accessed October 26, 2026.

[2] R. Sigrist, OEIS A382360, https://oeis.org/A382360, accessed October 26, 2026.